

23INMCA334–VISUAL PROGRAMMING LAB

Microproject Report Submitted By

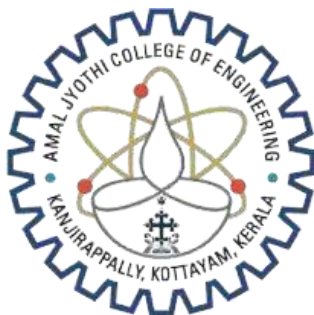
NANDANA PRAMOD

AJC23MCA-I048

In Partial fulfilment for the Award of the Degree Of

**INTEGRATED MASTER OF COMPUTER APPLICATIONS
(INMCA)**

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY



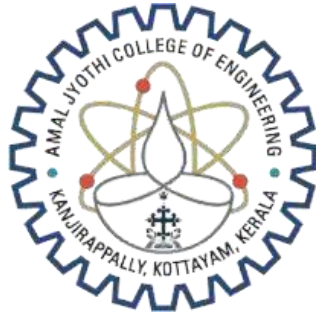
AMAL JYOTHI COLLEGE OF ENGINEERING AUTONOMOUS

KANJIRAPPALLY

[Affiliated to APJ Abdul Kalam Technological University, Kerala. Approved by AICTE,
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2025-2026

**DEPARTMENT OF COMPUTER APPLICATIONS AMAL JYOTHI
COLLEGE OF ENGINEERING AUTONOMOUS, KANJIRAPPALLY**



CERTIFICATE

This is to certify that the microproject report, “**23INMCA334 VISUAL PROGRAMMING LAB**” is the bonafide work of **NANDANA PRAMOD (AJC23MCA-I048)** in partial fulfillment of the requirements for the award of the Degree of Master of Computer Applications under Amal Jyothi College of Engineering Autonomous, Kanjirappally during the year **2025-26**.

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LIBNOVA

LIBRARY MANAGEMENT SYSTEM

Objective

The objective of this project is to develop a Library Management System that helps in managing books, members, issuing and returning books efficiently. It reduces manual work and ensures accurate record keeping.

System Description

LibNova is a VB.NET desktop application designed to simplify and automate library operations. It provides a secure login and signup system that allows users to access the application safely. The system includes complete book management features such as adding, updating, and deleting book records, making it easy to maintain an organized collection of books.

In addition to managing books, LibNova also supports member management, enabling the storage and maintenance of member details efficiently. The application allows users to issue books to members and record their return, ensuring proper tracking of library transactions. It also includes fine management, which helps in calculating and handling penalties for late returns.

A key feature of the system is the ability to search for books based on category, allowing users to quickly find the required books. The application is integrated with a SQL Server database, ensuring reliable data storage and retrieval. Overall, the system ensures fast, accurate, and efficient handling of all library operations.

Software & Hardware Requirements

The successful implementation of the LibNova Library Management System requires appropriate software and hardware resources to ensure smooth operation and efficient performance. These requirements support the development, execution, and database management of the system, enabling it to handle library activities such as book management, member management, and transactions effectively.

Software Requirements

- Visual Studio (VB.NET)
- SQL Server / LocalDB
- Windows Operating System

Hardware Requirements

- Processor: Intel i3 or above
- RAM: 4GB or more
- Storage: 500MB free space

Database Design

1. logintab

Column Name	Data Type	Description
reg_no	VARCHAR(20) (PK)	Registration number (Primary Key)
uname	VARCHAR(50)	Username
pword	VARCHAR(50)	Password

2. booktb

Column Name	Data Type	Description
bookid	INT (PK)	Book ID (Primary Key)
title	VARCHAR(100)	Book title
author	VARCHAR(100)	Author name
category	VARCHAR(50)	Book category

3. booktb

Column Name	Data Type	Description
bookid	INT (PK)	Book ID (Primary Key)
title	VARCHAR(100)	Book title
author	VARCHAR(100)	Author name
category	VARCHAR(50)	Book category

4. issuebktb

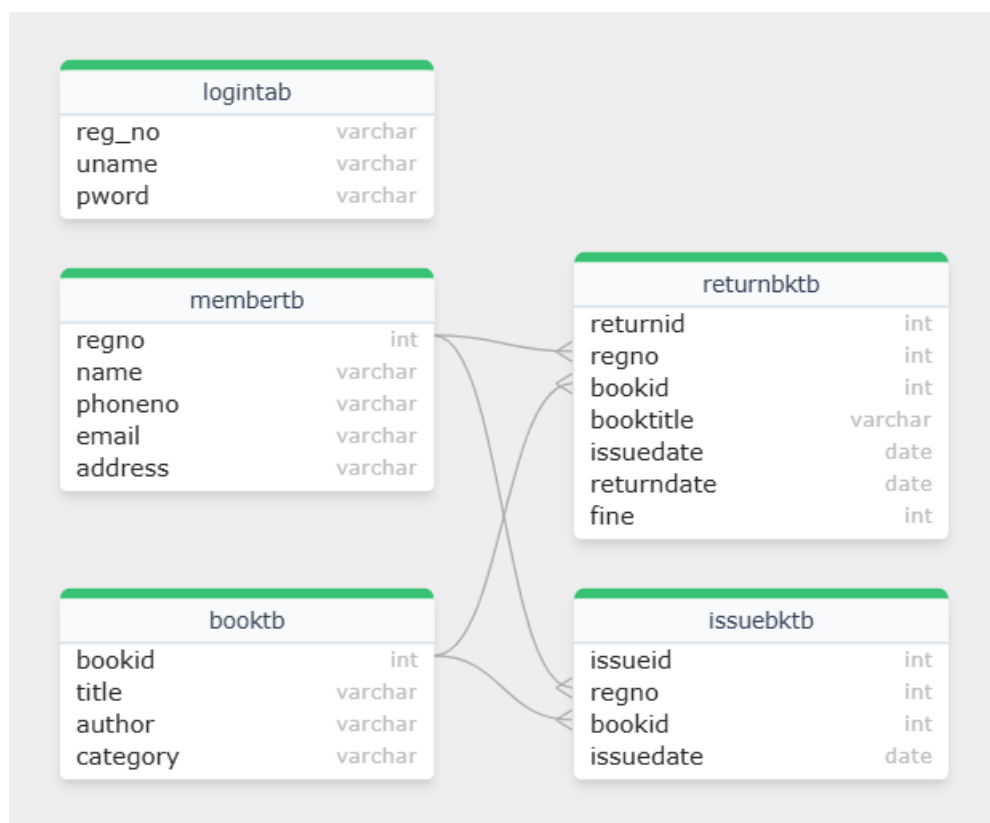
Column Name	Data Type	Description
issueid	INT (PK, Identity)	Issue ID (Auto increment)
regno	INT (FK)	References membertb(regno)
bookid	INT (FK)	References booktb(bookid)
issuedate	DATE	Date of issue

5. returnbktb

Column Name	Data Type	Description
returnid	INT (PK, Identity)	Return ID (Auto increment)
regno	INT (FK)	References membertb(regno)
bookid	INT (FK)	References booktb(bookid)
booktitle	VARCHAR(100)	Title of the book
issuedate	DATE	Date of issue
returndate	DATE	Date of return
fine	INT	Fine amount

ER Diagram

The Entity–Relationship (ER) diagram for the Library Management System provides a clear visual representation of the database structure and the relationships between different entities. It illustrates how data is organized into tables such as logintab, booktb, membertb, issuebktb, and returnbktb, each representing a specific component of the system. The diagram highlights key attributes like primary keys and foreign keys, showing how entities are connected—for example, linking members and books through issue and return transactions. Overall, the ER diagram helps in understanding data flow, maintaining data integrity, and designing an efficient and well-structured database system.



Form Design Explanation

The Library Management System is designed with multiple user-friendly forms to efficiently manage various operations such as user authentication, book handling, and member

management. Each form serves a specific purpose and is interconnected to ensure smooth workflow within the system. The design focuses on simplicity, data accuracy, and ease of navigation, enabling users to perform tasks like adding books, issuing and returning books, and searching records quickly. Special features like category-based book search and data display using DataGridView enhance usability and improve overall system functionality.

1. Login Form

- Validates username and password
- Redirects user to Main Menu after successful login

2. Signup Form

- Allows new users to register
- Stores user details in the database

3. Main Menu Form

- Contains navigation buttons
- Provides access to all modules of the system

4. Book Management Form

- Add, update, and delete book records
- Includes a **Display button** to trigger search functionality
- On clicking Display, a search panel appears
- User selects category from ComboBox
- Based on selection, books are filtered
- Results are shown in DataGridView

5. Member Management Form

- Add, update, and delete member details
- Stores member information in the database

6. Issue Book Form

- Issues books to members
- Records issue date and details

7. Return Book Form

- Handles book return process

- Calculates fine for late return
- Updates return records in database

Key Code Snippets Explanation

1. Insert Book

`INSERT INTO booktb VALUES(...)`

This SQL query is used to add a new book record into the booktb table. The values inside the brackets represent details such as book ID, title, author, and category. When this query is executed from the Book Management Form, it stores the entered book information into the database permanently.

2. Login Check

`SELECT COUNT(*) FROM logintab WHERE uname=@u AND pword=@p`

This query is used to validate user login credentials. It checks whether a record exists in the logintab table with the given username (@u) and password (@p). If the count returned is 1, it means the login is valid; if 0, the login fails. This ensures only authorized users can access the system.

3. Search by Category

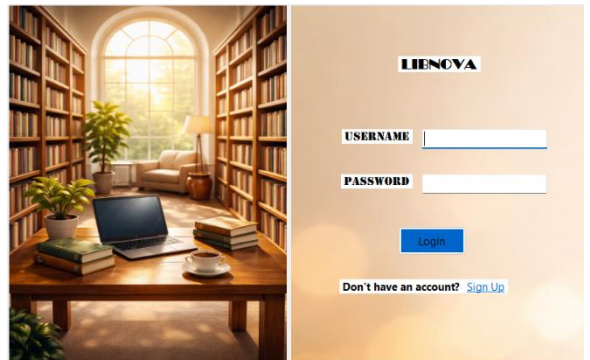
`SELECT * FROM booktb WHERE category=@category`

This query is used to retrieve books based on a selected category. The value @category is taken from the ComboBox selection in the Book Management Form. When executed, it filters and displays only the books that belong to the chosen category in the DataGridView, making it easier for users to find specific types of books.

Screenshots



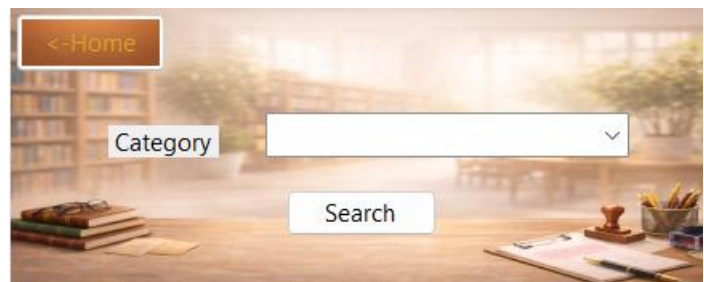
Front Page



Login Form



Main Menu Form



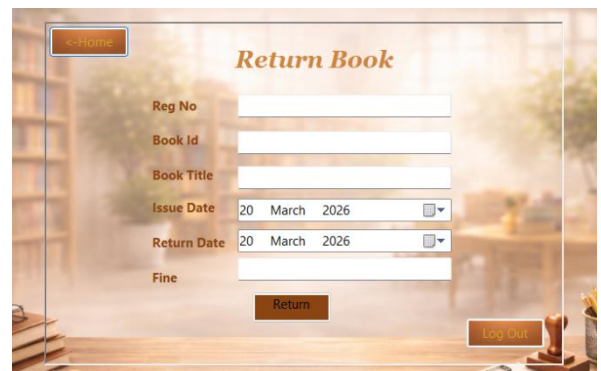
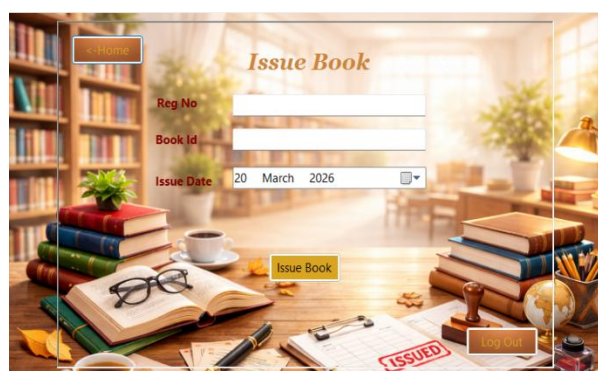
Search



Booking Mangement Form



Member Mangement Form



Issue Book Form

Return Book Form

Conclusion

The LibNova Library Management System successfully demonstrates an efficient and user-friendly approach to managing library operations. The system integrates essential functionalities such as book management, member management, issuing and returning of books, and fine handling, all supported by a reliable SQL Server database. The inclusion of features like category-based book search enhances usability and improves user experience. By automating manual processes, the system reduces errors, saves time, and ensures accurate record keeping. Overall, this project provides a practical solution for modern library management and can be further enhanced with additional features in the future.